Analytical Chemistry
Cumulative Exam
Spring 2015 4/24/15

1. (20 pts) Consider a beaker containing 10 mL of 0.15 F acetic acid (pKa = 4.75).
   a) Calculate the pH of the solution, using approximate equations below.
   b) Calculate the pH after adding to the beaker 5 mL of 0.1 M NaOH to the beaker.

2. (30 pts) Consider a beaker containing 10 mL of 0.15 F trisodium citrate, (pK_{a1} = 3.128, pK_{a2} = 4.76, pK_{a3} = 6.396)
   a) Abbreviate the salt as Na_3Ct, and write the K_{b1}, K_{b2}, and K_{b3} reactions.
   b) Calculate the numerical values for K_{b1}, K_{b2}, and K_{b3} and indicate the dominate reaction.
   c) Estimate the pH of the solution, using one of the formulas attached.
   d) Calculate the pH after adding to the beaker 5 mL of 0.1 M HCl to the beaker.

3. (40 pts) Just consider Na_3Ct in water. For problem 2a, you wrote the reactions pertaining to trisodium citrate, use these to write:
   a) The equilibrium expressions for each reaction
   b) Charge Balance (don't forget sodium) and
   c) Mass Balance for sodium citrate

4. (10 pts) a) Ignore the K_{b2} and K_{b3} reactions and using only the K_{b1} equilibrium rewrite the charge and material balance.
   b) Correctly, derive (DO NOT SOLVE) an equation that could be used to find the pH of this system. Your final equation may have $[OH^-]$ and as variables. If you did it right, your equation will reduce to

$$[OH^-] = \sqrt{[Ct^{3-}]K_{b1}}$$

Equations

$$pH = pK_a + \log\frac{f_b}{f_a}$$ or $$Ka = \frac{[H^+]fb}{fa}$$

$$[OH^-] = \frac{-K_b \pm \sqrt{K_b^2 + 4K_b f_b}}{2} \approx \sqrt{FbK_b}$$

$$[H^+] = \frac{-K_a \pm \sqrt{K_a^2 + 4K_a f_a}}{2} \approx \sqrt{FaKa}$$

$$pH + pOH = 14$$
$$Kw = 10^{-14} = [H^+][OH^-]$$
$$KaKb = Kw$$